

Amirhossein Yousefiramandi

+34 645 663 255 — Barcelona, Spain — [linkedin.com/in/amirhossein-yousefiramandi](https://www.linkedin.com/in/amirhossein-yousefiramandi) — github.com/amirhossein-yousefi — Google Scholar

Applied Scientist

Multimodal ML (Text↔Image) | Search & Ranking | Large-Scale Distributed | Publications + Patents

PROFESSIONAL SUMMARY — Applied scientist with 5+ years building production ML for search, ranking, and **multimodal text↔image** understanding at global scale. Trained CLIP/SigLIP retrieval reaching **R@1 90.7%** on Flickr8k (Img→Text); fine-tuned VLMs (**Qwen2-VL**, **PaliGemma2**) for multi-label classification via JSON generation; fine-tuned diffusion models for image generation / editing. First-author on two arXiv preprints (ModernBERT for patents; causal-LLM classification) and sole-author on two 2026 preprints; **three patents** at Clarivate. Stack: PyTorch, AWS SageMaker, Spark / Databricks, ONNX; languages: Python, Java, C++.

PROFESSIONAL EXPERIENCE

Clarivate — Intellectual Property Division

Aug. 2023 – Present

Senior Data Scientist (Applied Science)

Barcelona, Spain · hybrid

Clarivate is a global leader in analytics and information services across the IP, life sciences, and academia domains. The Intellectual Property Division operates the world's largest patent corpus and the search/analytics products that examiners, attorneys, and researchers use daily.

I joined the IP Division to drive the modernization of patent search, classification, and novelty workflows with foundation models. The role spans applied research — pretraining, fine-tuning, ablation design, and publication — and production ML engineering on AWS.

Domain: Product Delivery & Business Impact

- Designed and shipped the core ML classifier behind **Derwent AI Classifier** (public product) — **\$10M+ in incremental revenue** and a **+30% increase in customer renewal contracts**.

Domain: Domain-Specific Language Model Pretraining

- Pretrained three masked LMs from scratch on a **60M+ patent corpus** (**ModernBERT** architecture, custom BPE tokenizer, FlashAttention, RoPE, GLU FFNs); **ModernBERT-base-PT** outperforms general ModernBERT on 3/4 internal benchmarks at **>3×** the inference throughput of PatentBERT.
- Designed advanced cleaning and deduplication pipelines for the patent corpus; ran controlled studies on data mixtures, tokenizer extension, and CUDA optimizations.
- Published as: **Patent Language Model Pretraining with ModernBERT**, arXiv:2509.14926 (first author).

Domain: Patent Search & Classification

- Spearheaded end-to-end design, training, and deployment of two proprietary ML classifiers for patent search and classification; owned research-to-production delivery across teams.
- Designed hybrid **LLM/BERT taxonomy-mapping** and query-understanding services on AWS, combining per-level supervised models with prompt-based labeling and hierarchical inference — powering discovery and navigation use cases.
- Delivered an unsupervised, real-time hybrid **semantic search system** for portfolio benchmarking, with large-scale preprocessing and feature engineering in PySpark / Databricks.
- Co-author of a study comparing **embedding-based vs. instruction-based** fine-tuning of decoder-only LLMs under 4-bit + LoRA on a single GPU (arXiv:2512.12677).
- Both classifier systems protected by **pending USPTO applications** (App. 19/029,628; App. 18/590,188 / Pub. US20250272608A1).

Domain: Multi-Agent Novelty & Prior-Art Research

- **Business impact:** the capability **secured engagement from several major enterprise customers** and **materially lifted our new-contract acquisition rate**.
- Designed and shipped a multi-agent novelty / prior-art research system on **AWS AgentCore**: specialized agents call semantic patent search, keyword search, and NPL (scientific literature) search tools.
- Deployed with input/output guardrails against **prompt injection** and data leakage.
- Engineered Reflexion-style agents with **LangGraph** to orchestrate retrieval, ranking, and LLM reasoning over large patent corpora for interactive analysis.

Domain: Generative AI & Diffusion

- Ran controlled comparisons of **SFT / LoRA / QLoRA / continued pretraining** across Llama and Gemma variants; deployed best configurations via AWS DLC and MCP servers.
- Integrated with **Amazon Bedrock, Google Gemini, and Azure AI**.
- Trained and fine-tuned **diffusion models** for image generation and editing: dataset curation, conditioning design, training stabilization, inference optimization.
- Built **interpretability tooling** (attention attribution, contrastive explanations) for transformer models to meet transparency / traceability requirements for enterprise customers.

Domain: MLOps & Production Infrastructure

- Owned the lifecycle of ML services and data pipelines for high-volume workloads on AWS (**SageMaker, Glue, Fargate**); focused on scalable, low-latency inference and cost-efficient compute.
- Orchestrated labeling and evaluation workflows on **AWS Step Functions** (Lambda / ECS / SageMaker Processing) with fan-out/fan-in parallelism, retries / backoff, and failure handling.
- Developed reusable speech / audio pipelines (**ASR, TTS, KWS, voice conversion**) with ONNX export and SageMaker real-time / serverless / batch-transform deployment.
- Drove **engineering excellence** across the ML stack — owned end-to-end development and deployment, led code reviews, built **CI/CD pipelines** and automated test suites, and championed **clean-code** / type-checking standards on the production codebase.

Double Avatars

Feb. 2023 – Aug. 2023

AI Engineer

Spain · on-site

Double Avatars develops virtual try-on and 3D avatar experiences for e-commerce platforms.

- Developed **virtual try-on** and **3D avatarization** models combining generative image models with body-shape estimation.
- Streamlined end-to-end training and inference pipelines for computer vision models, increasing reliability and runtime efficiency.
- Collaborated with UX/UI and backend teams to integrate ML services into production and iterate via experimentation.

Freelance

Sep. 2022 – Feb. 2023

Freelance Data Scientist

Spain

- Enhanced **semantic search engines**, improving **MRR@15 by 8%** via iterative model and ranking-feature improvements.
- Built multimodal **price-estimation and recommender systems**, reducing **MAE by 12%** by combining vision and text signals with robust evaluation.
- Mentored **20+ ML students** through structured projects, code reviews, and deployment best practices.

Semrush

May 2022 – Aug. 2022

Data Scientist

Spain

- Developed **active-learning NLP pipelines** on large-scale web text to accelerate high-quality data labeling.
- Built and deployed NLP models to extract business insights from unstructured text.

CafeBazaar

Nov. 2020 – May 2022

Data Scientist

Tehran, Iran

CafeBazaar is Iran's largest mobile application marketplace.

- Implemented and productionized **semantic app-search** models, increasing **Recall@20 by 9%** and improving ranking quality for a large-scale marketplace.
- Built automated **POI extraction** pipelines with YOLO, OCR, and SATRN; **F1 53 → 65%** while halving inference time.
- Optimized **multi-GPU training** for segmentation, achieving **+11% mIoU** on vehicle-pricing use cases and reducing time-to-train.

- Developed activity-recognition systems that significantly reduced error rates in behavioral-understanding features.
- *Twice promoted within one year for outstanding performance.*

Baharan
Data Scientist

Feb. 2020 – Oct. 2020
Iran

- Developed **facial analysis / recognition** models for surveillance systems.
- Shipped new **facial-attribute detection** features that expanded real-time use cases.

TECHNICAL SKILLS

LLMs & NLP

- Architectures** Transformer internals (MHA, GQA, MoE, MLA), attention variants (Flash, paged, grouped-query), positional encodings (RoPE, ALiBi)
- Training** SFT, RLHF, DPO, GRPO, LoRA / QLoRA, knowledge distillation, continued pretraining, tokenizer extension
- Inference** KV-cache reuse, speculative decoding, 4/8-bit quantization, ONNX export, batch packing
- Toolkits** UnSloth (Llama / Gemma), MosaicML Composer, TRL, PEFT, vLLM, HF Transformers

Multimodal / Vision / Speech

- Vision-Language** CLIP, SigLIP, Qwen2-VL, PaliGemma2, SmolVLM
- Speech / Audio** Whisper, Wav2Vec2 (CTC), SpeechT5, WavLM, ECAPA, HiFi-GAN
- Image** Diffusers (fine-tuning, conditioning), YOLO, OCR, SATRN

Agents & Retrieval

- Agents** LangGraph, LlamaIndex, MCP, Strands Agents, AgentCore, Agent Development Kit
- Retrieval** dense retrieval, BM25, hybrid + RRF, FAISS, Elasticsearch, LLM-as-judge evaluation

MLOps & Cloud

- AWS** SageMaker (real-time, serverless, batch transform, HPO), Glue, Lambda, Fargate, Step Functions, Bedrock, DLC
- Other cloud** Azure AI, Google Gemini integrations
- Data** Spark / Databricks (PySpark), SQL
- Infra** Docker, Git, MLflow, ONNX Runtime, $\LaTeX$

Languages & Methods

- Languages** Python, C++, Bash, SQL
- Quantitative** mathematical optimization, statistical learning, information theory

SELECTED OPEN-SOURCE PROJECTS

Text-JEPA Experiments & AF-JEPA (PyTorch, MiniLM, Qwen 2.5-7B)

- Two empirical studies on JEPA target design for text: semantic-card target ablation across 4 seeds (vs. raw embeddings / summaries / shuffled controls) on WikiText-2-raw, and **AF-JEPA** (frozen-LLM target encoder, smooth-L1 alignment on FineWeb-Edu).
- Block-masking (I-JEPA style) + InfoNCE / cosine alignment; raw target reaches 0.242 ± 0.016 top-1 retrieval vs. semantic cards 0.183 ± 0.008 . GitHub.

Multilingual LLM Adaptation via MAD-X-Style Adapter Stacking (Swahili Biomedical)

- Composable adapter stack on a frozen **AfroXLMR-large** backbone: Swahili language adapter + English biomedical domain adapter (MLM-trained) + supervised task adapters, stacked at inference (MAD-X / Pfeiffer adapters).
- Evaluated on medical MCQA (**MMLU-ProX-Swahili clinical**), Swahili NER (MasakhaNER, BC5CDR-Disease projected), and topic classification; training data from NLLB-200 and OpenWHO. GitHub.

Instruction-Tuned LLM for Text Classification (LoRA/QLoRA)

- TRL, PEFT, decoder-only LLM outputting exact label strings; demo ~**94%** accuracy. GitHub · Model.
Multi-Modal Hateful Content Classification (CLIP/SigLIP Fusion + MTL)
- Late-fusion gated attention + per-label MTL on MMHS150K; **macro-F1 0.569, ROC-AUC (macro) 0.783**.
GitHub.
Text+Image VLM Fine-Tuning via JSON Generation (MMHS150K)
- Generative classification with Qwen2-VL / PaliGemma2 / SmolVLM; Qwen2-VL+LoRA **micro-F1 0.611** (test).
GitHub.
Multi-Label Emotion Classification (GoEmotions / Reddit)
- RoBERTa-base 27-emotion + neutral; **micro-F1 0.528 @ thr 0.84, ROC-AUC 0.952**. GitHub.
Whisper Multi-Task ASR + Voice Intensity
- Joint speech recognition + loudness regression on Whisper encoder; **WER 4.6976%**, intensity **RMSE 0.7334**;
Gradio demo + SageMaker helpers. GitHub.
Wav2Vec2 Multi-Task: CTC ASR + Intensity
- Wav2Vec2-base-960h (CTC) + intensity head with attention-masked mean pooling; validation **WER ≈12.9%**;
config-first repo, SageMaker-packaged. GitHub.
CTC ASR Baseline (Speech-to-Text)
- End-to-end Wav2Vec2 pipeline (Common Voice 17.0) with robust audio I/O, text normalization, ONNX export,
CLI inference, and SageMaker autoscaling / HPO. GitHub.
Emotion-Aware TTS Style Transfer
- SpeechT5 + WavLM + ECAPA + HiFi-GAN: StyleAdaptor and StyleSpeakerFusion; Gradio UI, optional
PEFT/LoRA fine-tuning. GitHub.
Keyword Spotting (Wav2Vec2, ONNX, SageMaker)
- Real-time streaming inference with sliding-window smoothing and offline batch scoring on Speech Commands
v2; SageMaker realtime / serverless / batch helpers. GitHub.
Any-to-Any Speech Conversion (Prototype)
- SpeechT5-VC + ECAPA + HiFi-GAN: convert source utterance to target speaker's timbre using a short reference
clip; CMU ARCTIC fine-tuning loop, hosted-inference handlers. GitHub.
Image-Text Retrieval (CLIP/SigLIP)
- Contrastive training and FAISS-ready embeddings; Flickr8k **R@1 90.7%** (Img→Text), **77.06%** (Text→Img);
Flickr30k **R@1 92.3% / 79.0%**. GitHub.
Forward-Forward Layer Free-Riding Analysis
- Companion code for arXiv:2605.06240 (sole author); per-block, hardness-gated, and depth-scaled remedies on
CIFAR-10/100 and Tiny ImageNet.

PUBLICATIONS AND PATENTS

Patents

- **Camelot Docket No. I03.00450000 — Prior Art Search Using Artificial Intelligence Agents** (pending — in
verification, 2026). Assignee: Camelot UK Bidco Ltd. (Clarivate). **Sole inventor: Yousefiramandi, A.** Covers
the multi-agent AgentCore prior-art / novelty-search system deployed at Clarivate.
- **U.S. Patent App. No. 19/029,628 / Camelot Docket No. I03.00430000 — Training an Artificial Intelligence
Classifier Using LLM-Labeled Training Data** (pending; signed 2025-01-14). Assignee: Camelot UK Bidco Ltd.
First inventor: Yousefiramandi, A.
- **U.S. Patent App. No. 18/590,188 / Pub. US20250272608A1 — Hybrid Artificial Intelligence Classifier**;
Mishra, S., Cooney, C., **Yousefiramandi, A.**, & Mishra, A. (2025). patents.google.com. *Co-inventor*.
Peer-reviewed / Preprint Papers
- **Yousefiramandi, A. & Cooney, C. (2025). Patent Language Model Pretraining with ModernBERT.**
arXiv:2509.14926. **[first author, 4 citations]**
- **Yousefiramandi, A. & Cooney, C. (2025). Fine-Tuning Causal LLMs for Text Classification: Embedding-Based
vs. Instruction-Based Approaches.** arXiv:2512.12677. **[first author]**

- Yousefiramandi, A. (2026). *Cumulative-Goodness Free-Riding in Forward-Forward Networks: Real, Repairable, but Not Accuracy-Dominant*. arXiv:2605.06240. [sole author]
- Yousefiramandi, A. (2026). *Learning Selective Merge Policies for Deadline-Constrained Coded Caching via Deep Reinforcement Learning*. arXiv:2605.15236. [sole author] — GAT policy network trained with PPO; 40.9% reduction in broadcast-packet expiration ratio over baselines.

EDUCATION

Sharif University of Technology	2018 – 2021
<i>M.Sc. in Electrical Engineering (AI)</i>	<i>Tehran, Iran</i>
Isfahan University of Technology	2014 – 2018
<i>B.Sc. in Electrical Engineering</i>	<i>Isfahan, Iran</i>

HONORS AND AWARDS

- Promoted at **Clarivate** within less than a year for exceptional performance.
- **Three patents** (**sole inventor** on prior-art AI agents; **first inventor** on LLM-labeled training data; co-inventor on hybrid AI classifier) and two **papers** (ModernBERT; causal-LLM classification) authored during the Clarivate tenure, plus two sole-author 2026 preprints (Forward-Forward; DRL coded caching).
- **Twice promoted** at CafeBazaar within one year.
- Ranked **16th nationwide** in Iran’s master’s-program entrance examination.

LANGUAGES

- **English** — fluent (professional working proficiency)
- **Spanish** — working proficiency
- **Persian** — native